CIS 635: Connecting to the GVSU Clipper HPC Cluster

Student instructions · Fall 2026

Overview

Your CIS 635 section now has access to Clipper, GVSU's high-performance computing (HPC) cluster. Please complete the connection test in this guide before class so any access problems surface early.

There are two ways into Clipper:

- **OpenOnDemand (web browser):** deck.clipper.gvsu.edu. This is the easiest route and the one we will use for Jupyter.
- **SSH (terminal):** `ssh username@clipper.gvsu.edu`, for command-line work and batch jobs.

Both routes work on campus wired networks and through the GlobalProtect VPN. **Connect to the VPN first in every case** (Step 1). On student Wi-Fi or off campus, Clipper is not reachable without it.

Timing note from GVSU IT: access was just granted to our sections. It can take a few hours for the change to sync to Okta, so if OpenOnDemand rejects you on the first day, wait a few hours and try again before reporting a problem.

Before you start

- **GVSU network account.** Your usual GVSU username and password (the username is the part before `@mail.gvsu.edu`).
- **Multi-factor authentication (MFA) enrolled.** The VPN requires it. If you have never set it up, sign in at my.gvsu.edu and follow the enrollment prompts (Okta Verify is recommended). See MFA for Students.
- **Admin rights on your laptop** to install the VPN client.
- **An SSH client.** macOS and Linux: the built-in Terminal. Windows 10 (1809 or later) and 11: PowerShell has `ssh` built in. PuTTY also works.

You do **not** need Duo for SSH as long as you are connected to the VPN, because the VPN already handles two-factor authentication.

Step 1: Install and connect the GlobalProtect VPN

GVSU's VPN is GlobalProtect. The portal address is the same on every platform: **`gvsu.gpcloudservice.com`**. Full illustrated steps are in the BYOD GlobalProtect install guide; the short version is below. GVSU's guide assumes you install while on campus Wi-Fi.

Windows

1. Download the installer from the BYOD guide. Use the “Microsoft Surface” version if your laptop has an ARM processor (e.g., Surface Pro 11, Snapdragon laptops).
2. Run it from your Downloads folder, click **Next** through the installer, approve the User Account Control prompt, then **Close**.
3. Restart your computer.
4. Click the GlobalProtect globe icon in the system tray (under the “Show hidden icons” arrow).
5. Enter `gvsu.gpcloudservice.com` as the portal, click **Connect**, and sign in with your GVSU credentials and MFA.

macOS

1. Download the `.pkg` installer from the BYOD guide and open it.
2. Click **Continue**, **Continue**, **Install**, enter your Mac password, then **Close**.
3. Click the GlobalProtect globe icon in the menu bar, enter `gvsu.gpcloudservice.com`, and click **Connect**.

Linux (Ubuntu)

1. Download the `.tgz` from the BYOD guide and extract it.
2. Run `gp_install.sh` and enter your password when prompted.
3. When GlobalProtect opens, enter `gvsu.gpcloudservice.com` and click **Connect**.

Check: the GlobalProtect window shows **Connected** and a gateway name (e.g., US Central). If it says “You are on the internal corporate network,” you are on a campus network; on student Wi-Fi, use the **Change Gateway** option to pick **US Central** (VPN overview).

If sign-in fails with an error right after you enter your password, your account is most likely not enrolled in MFA. Enroll at my.gvsu.edu, then try again.

Step 2: Test the OpenOnDemand web login

1. With the VPN connected, open deck.clipper.gvsu.edu in your browser.
2. You are redirected to Okta. Sign in with your GVSU credentials.
3. You should land on the OpenOnDemand dashboard, with menus for **Files**, **Jobs**, **Clusters** and **Interactive Apps**.
4. Open **Clusters → Clipper Shell Access** (the exact label may differ slightly). A terminal opens in a new tab; type `hostname` and press Enter. Seeing a reply confirms your account works on the cluster.
5. Open **Files → Home Directory**. This file browser is how you will upload datasets and notebooks and download results.

Pass: you see the dashboard and can open your home directory. Take a screenshot of the dashboard for the check-in (see the checklist at the end).

Step 3: Test the SSH connection

Clipper's address is `clipper.gvsu.edu` (also `port.clipper.gvsu.edu`). Both connect you to **Port**, the login node from which all jobs are submitted (Accessing Clipper).

1. Confirm the VPN shows **Connected**.
2. Open Terminal (macOS/Linux) or PowerShell (Windows) and run, with your own username:

```
ssh username@clipper.gvsu.edu
```

3. The first time, you are asked whether to trust the host. Check that it says `clipper.gvsu.edu`, type `yes`, and press Enter.
4. Enter your GVSU password. Nothing appears as you type; that is normal. With the VPN on, you should not get a Duo prompt.
5. Run these checks and confirm each gives a sensible answer:

```
whoami                # your GVSU username
hostname              # the login node's name
pwd                   # your home directory
python3 --version     # system Python (3.9)
module avail python   # newer Python versions available as modules
sinfo -p class        # the class partition you are allowed to use
```

6. Type `exit` to log out.

Important: the login node is shared by everyone. Do not run heavy computation directly on it. Use Jupyter (next section) or a batch job on the `class` partition instead.

Your resource limits

Students can submit only to the `class` partition. Per student, at one time (HPC Classroom Usage):

Resource	Maximum per student
CPU cores	4
Memory	32 GB
GPU shards (a slice of a shared GPU)	4

A job asking for more is rejected. Several jobs from you together cannot exceed these limits, so extra jobs wait in the queue. Submitting to any other partition fails with `Invalid account or account/partition combination specified`.

Optional batch-job test: save the following as `test_job.sh`, run `sbatch test_job.sh`, check it with `squeue -u $USER`, and read `output_<jobid>.txt` when it finishes.

```
#!/bin/bash
#SBATCH --job-name=cis635_test
#SBATCH --output=output_%j.txt
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=2G
#SBATCH --partition=class

echo "Hello from $(hostname)"
python3 --version
```

Running Jupyter notebooks on Clipper

Jupyter runs on Clipper through OpenOnDemand's **Jupyter Server** app, which starts a JupyterLab session on a compute node and opens it in your browser (Accessing JupyterLab on Clipper). The notebooks you run in Colab will open here unchanged.

Part A (one time): create a Python environment for the course

GVSU recommends a Python virtual environment for each project (Python Environments on Clipper). Do this once, from an SSH session or the OpenOnDemand shell:

```
cd ~
module load python/3.12   # optional; default is 3.9. Check names: module avail python
python3 -m venv cis635
source cis635/bin/activate
python3 -m pip install --upgrade pip
python3 -m pip install ipykernel numpy pandas scikit-learn matplotlib seaborn plotly
python3 -m ipykernel install --user --name cis635 --display-name "Python (cis635)"
deactivate
```

The last install line registers the environment so it appears as a kernel named **Python (cis635)** in JupyterLab. Install any further packages the same way: activate, `pip install`, deactivate. If you prefer conda, run `module load miniconda3` first and follow the conda steps in the same GVSU article.

Part B (each time): launch a Jupyter session

1. Connect the VPN and sign in at deck.clipper.gvsu.edu.
2. Open **Interactive Apps → Jupyter Server**.
3. Fill in the session form. Stay within the class limits: at most **4 cores** and **32 GB** memory. Request only the hours you need (for example 2), since the session ends automatically when the time runs

out. Leave the GPU option off unless an assignment needs one; GPU sessions from OpenOnDemand automatically get 4 shards.

4. Click **Launch**. The session is queued and becomes ready, usually within seconds. If all class resources are busy, it waits in the queue.
5. Click **Connect to Jupyter Server**. JupyterLab opens in a new tab.
6. Create or open a notebook and choose the **Python (cis635)** kernel (**Kernel → Change Kernel** for an existing notebook).
7. Run this cell to confirm everything works:

```
import sys, platform, os
import numpy, pandas, sklearn
print("Python:", sys.version.split()[0])
print("Node:", platform.node())
print("Cores available:", len(os.sched_getaffinity(0)))
print("pandas", pandas.__version__, "| scikit-learn", sklearn.__version__)
```

The node name should differ from the login node you saw over SSH, which shows the notebook is running on a compute node.

Moving files and ending the session

- **Upload/download:** use **Files → Home Directory** on the dashboard, or the upload button in JupyterLab's file panel. From a terminal, `scp mydata.csv username@clipper.gvsu.edu:~/` also works while on the VPN.
- **Colab differences:** remove Colab-only lines such as `from google.colab import drive`; read data from your Clipper home directory instead. `!git clone` and `!pip install` work, but installing into your `cis635` environment from the terminal is more reliable.
- **When you finish:** save, then go to **My Interactive Sessions** on the dashboard and click **Delete** to free your cores for classmates. Closing the browser tab alone keeps the session running until its time limit; you can reconnect from the same page.

Troubleshooting

Symptom	Likely cause	What to do
OpenOnDemand login refused or “not authorized” on the first day	Okta has not synced the new access yet	Wait a few hours and retry
VPN sign-in error right after password	MFA not enrolled	Enroll at my.gvsu.edu, sign out fully, retry
`ssh` hangs or “connection timed out”	VPN not connected (or on student Wi-Fi without VPN)	Connect GlobalProtect, confirm **Connected**, retry
SSH asks for Duo	VPN not connected	Connect the VPN; Duo is bypassed on VPN
“Permission denied” over SSH	Wrong username or password	Use your GVSU username only (no `@mail.gvsu.edu`)
`Invalid account or account/partition combination specified`	Job sent to a partition other than `class`	Add `#SBATCH --partition=class`
Job rejected, or Jupyter form error on launch	Requested more than 4 cores / 32 GB / 4 shards	Lower the request
Jupyter session stuck at “Queued”	All class resources in use	Wait, or delete your own older sessions
Kernel **Python (cis635)** missing	Environment not registered	Rerun the `ipykernel install` line in Part A
`ModuleNotFoundError` in a notebook	Wrong kernel or package not installed	Switch to **Python (cis635)**; install the package into that environment

If none of these helps, email me with a screenshot of the error, your operating system, and whether the VPN showed Connected. GVSU IT has asked to hear about any issues, so I will forward problems I cannot solve.

Connection checklist

Submit screenshots of items 2, 3 and 5.

- [ ] 1. GlobalProtect VPN installed and shows Connected
- [ ] 2. OpenOnDemand dashboard loads at deck.clipper.gvsu.edu
- [ ] 3. SSH login works and `whoami` / `hostname` print correctly
- [ ] 4. `cis635` environment created and registered as a Jupyter kernel
- [ ] 5. Jupyter Server session launched and the test cell runs on the Python (cis635) kernel
- [ ] 6. Session deleted from My Interactive Sessions afterwards

GVSU references

- Accessing Clipper HPC Cluster
- GlobalProtect VPN Overview
- BYOD: Installing the GlobalProtect VPN Client
- HPC Classroom Usage
- Accessing JupyterLab on Clipper
- Creating and Using Python Environments on Clipper
- Using GPUs on Clipper
- Multi-Factor Authentication for Students